Hartnell Governors

by

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Spring Controlled Governor:

- It consists of two bell crank levers pivoted at the points (0,0) to the frame.
- The frame is attached to the governor spindle and therefore rotates with it.
- Each lever carries a ball at the end of the vertical arm OB and a roller at the end of the horizontal arm OR.
- A **helical compressive spring provides equal downward forces on the two rollers through a collar on the sleeve**.
- It serves the purpose like dead weight load.
- The **spring force may be adjusted by screwing a nut up or down on the sleeve**.

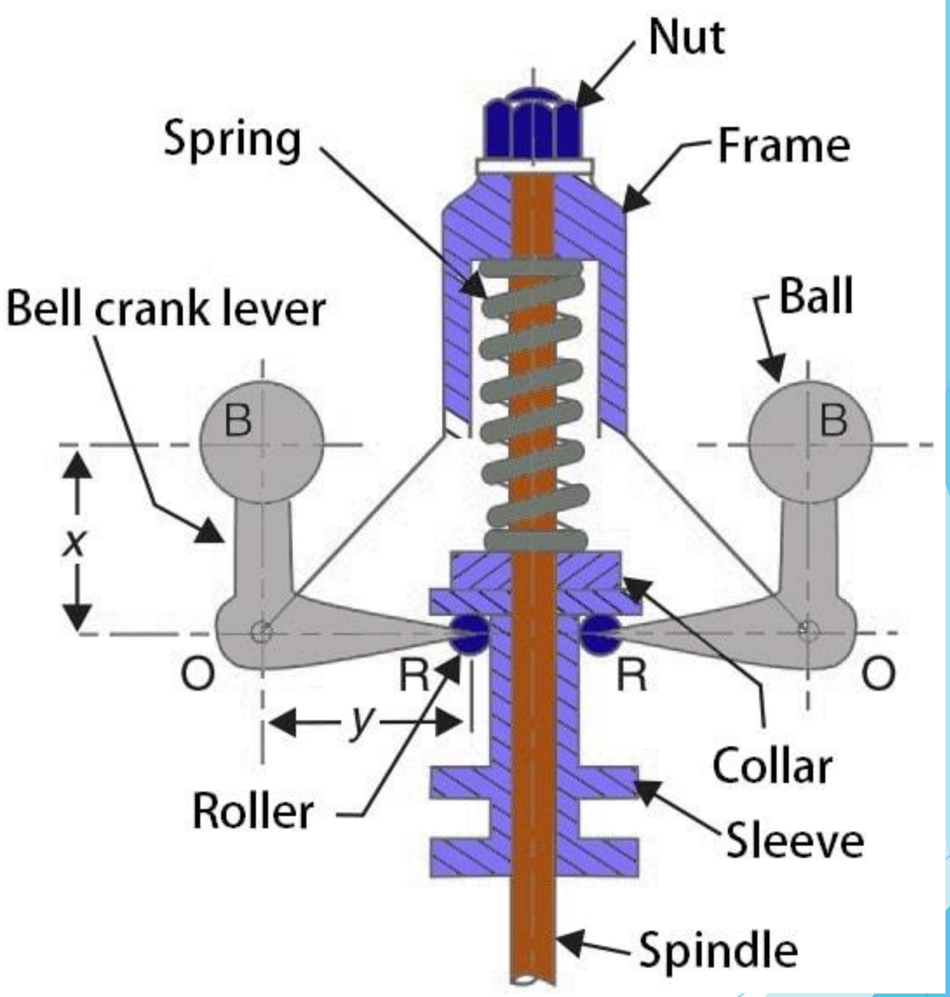


Fig. 1. Hartnell Governor

Reference: extrudesign.com

Spring Controlled Governor : Hartnell Governor contd…

m = Mass of each ball in kg

M = Mass of sleeve in kg

r_1 = Minimum radius of rotation in meter

r_2 = Maximum radius of rotation in meter

ω1 = Angular speed of the governor at minimum radius in rad/s

ω2 = Angular speed of the governor at maximum radius in rad/

*S*1 = Spring force exerted on the sleeve at ω1 in newtons

*S*2 = Spring force exerted on the sleeve at ω2 in newtons

F_{C1} = Centrifugal force at ω1 in newtons = $m\,(\omega_1)^2\,r_1$,

F_{C2} = Centrifugal force at ω2 in newtons = $m\,(\omega_2)^2\,r_2$,

s = Stiffness of the spring or the force required to compress the spring by one mm

x = Length of the vertical or ball arm of the lever in meter

y = Length of the horizontal or sleeve arm of the lever in meter

r = Distance of fulcrum *O* from the governor axis or the radius of rotation when the governor is in mid-position, in meter

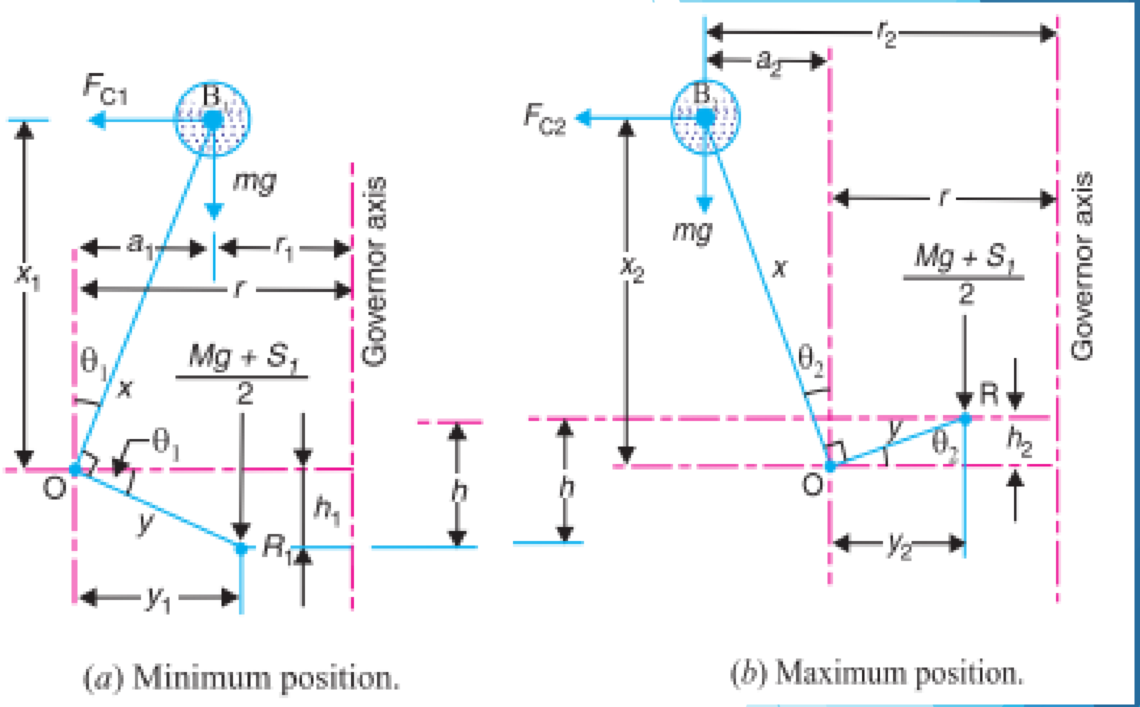


Fig. 2. Line diagram of Hartnell Governor

Reference: R.S. Khurmi and J.K. Gupta, Theory of Machines, S. Chand.

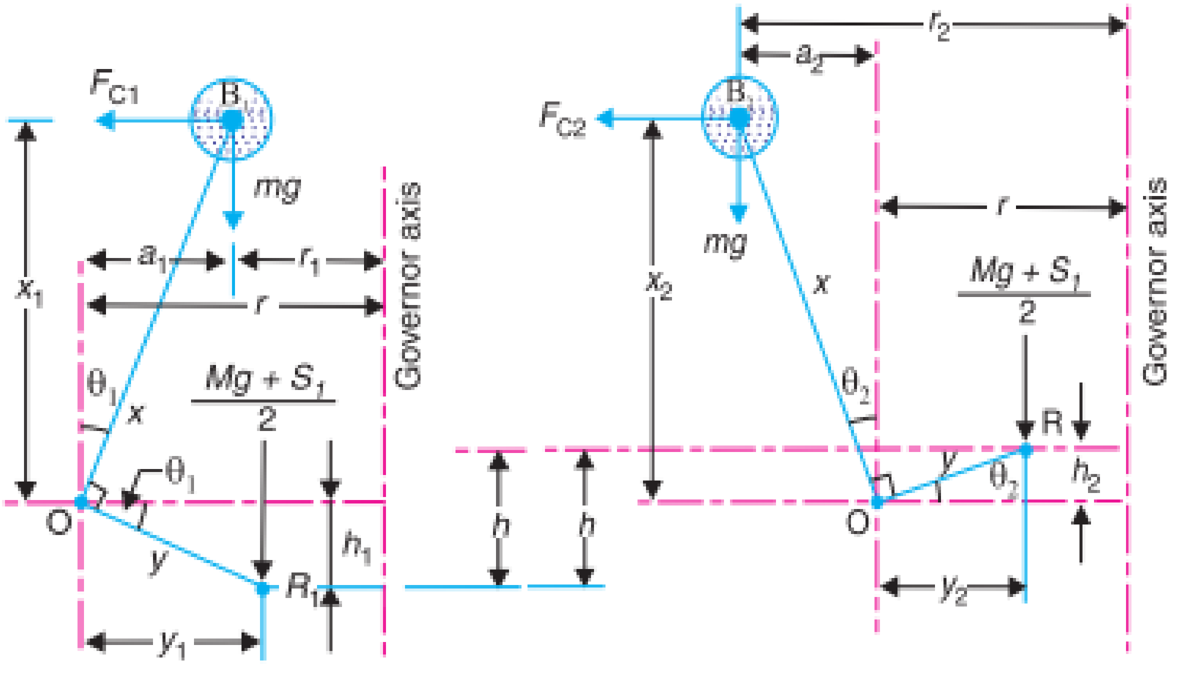
Hartnell Governor contd…

- For the minimum position *i.e.* when the radius of rotation changes from r to r_1, as shown in Fig. 3(a), the compression of the spring or the lift of sleeve h_1 is given by:

$$\frac{h_1}{y} = \frac{a_1}{x} = \frac{r - r_1}{x} \qquad (1)$$

- Similarly, for the maximum position *i.e.* when the radius of rotation changes from r to r_2, as shown in Fig. 3(b), the compression of the spring or lift of sleeve h_2 is given by

$$\frac{h_2}{y} = \frac{a_2}{x} = \frac{r_2 - r}{x} \qquad (2)$$



(*a*) Minimum position. (*b*) Maximum position.

Fig. 3. Line diagram

Hartnell Governor contd…

Adding equations **(1) and (2)**

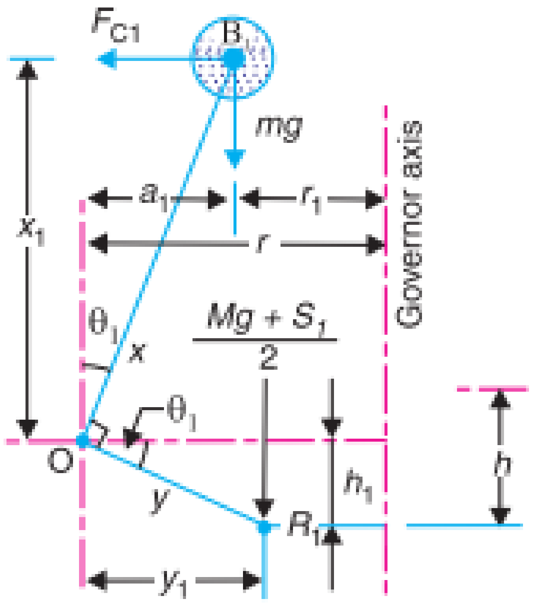
$$\frac{h_1 + h_2}{y} = \frac{r_2 - r_1}{x} \quad or \quad \frac{h}{y} = \frac{r_2 - r_1}{x}$$

$$Thus, \qquad h = (r_2 - r_1)\frac{y}{x} \qquad (3)$$

Now for minimum position, taking moments about point *O*, we get

$$F_{C1} \times x_1 = \frac{M.g + S_1}{2} \times y_1 + m.g \times a_1$$

$$M.g + S_1 = \frac{2}{y_1}\left(F_{C1} \times x_1 - m.g \times a_1\right) \qquad (4)$$



(a) Minimum position.

Fig. 4.

Hartnell Governor contd…

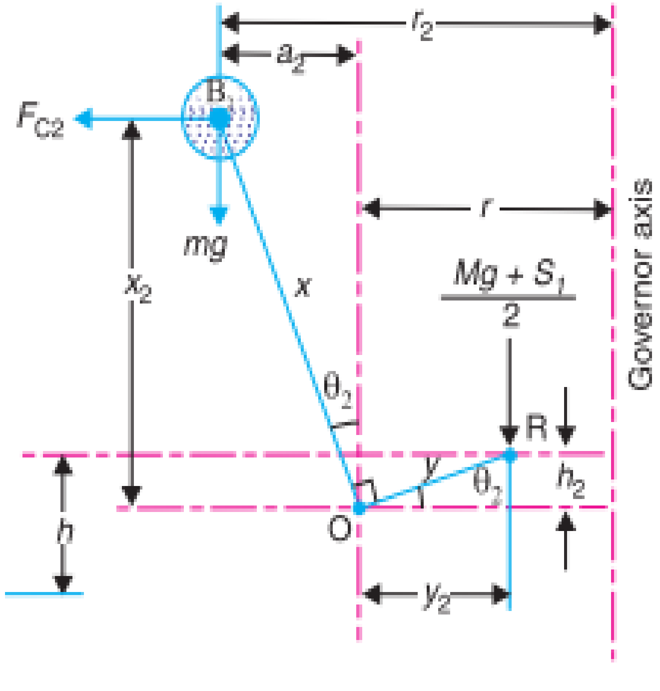
Again for maximum position, taking moments about point *O*, we get:

$$F_{C2} \times x_2 = \frac{M.g + S_2}{2} \times y_2 + m.g \times a_2$$

$$M.g + S_2 = \frac{2}{y_2}\left(F_{C2} \times x_2 - m.g \times a_2\right) \quad (5)$$

Subtracting equation **(4) from equation (5):**

$$S_2 - S_1 = \frac{2}{y_2}\left(F_{C2} \times x_2 - m.g \times a_2\right) - \frac{2}{y_1}\left(F_{C1} \times x_1 - m.g \times a_1\right)$$



(b) Maximum position.

Fig. 5.

In the working range of governor, θ **is usually small and so the obliquity effects of the arms of the bell-crank levers may be neglected.** **So, $x_1 = x_2 = x$, $y_1 = y_2 = y$ and $a_1 = a_2 = 0$**

Hartnell Governor contd…

Subtracting equation (4) from equation (5):

$$S_2 - S_1 = 2\left(F_{C2} - F_{C1}\right) \times \frac{x}{y} \qquad (6)$$

We know:

$$S_2 - S_1 = h.s \quad and \quad h = \left(r_2 - r_1\right)\frac{y}{x}$$

$$s = \frac{S_2 - S_1}{h} = \left(\frac{S_2 - S_1}{r_2 - r_1}\right)\frac{x}{y} \qquad (7)$$

Substituting equation (6) into equation (7), we get:

$$s = \frac{S_2 - S_1}{h} = 2\left(\frac{F_{C2} - F_{C1}}{r_2 - r_1}\right) \times \left(\frac{x}{y}\right)^2 \qquad (8)$$

- When friction is taken into account, the weight of the sleeve ($M.g$) **may be replaced by ($M.g. \pm F$).**
- **The + sign is used when the sleeve moves upwards or the governor speed increases** and **- sign is used when the sleeve moves downwards** or **the governor speed decreases.**

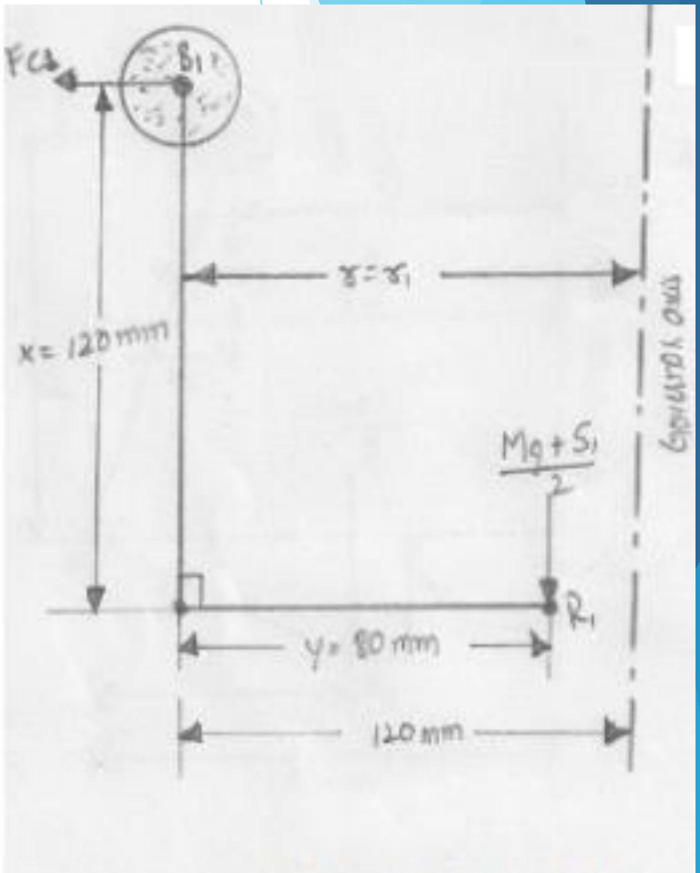
Problem 1. A Hartnell governor having a central sleeve spring and two right-angled bell crank levers moves between 290 rpm. and 310 rpm. for a sleeve lift of 15 mm. The sleeve arms and the ball arms are 80 mm and 120 mm, respectively. The levers are pivoted at 120 mm from the governor axis and mass of each ball is 2.5 kg. The ball arms are parallel to the governor axis at the lowest equilibrium speed. Determine:

1. Loads on the spring at the lowest and the highest equilibrium speeds
2. Stiffness of the spring

Solution. Given : *m* = 2.5 kg ; *r* = 120 mm = 0.12 m; h = 15 mm = 0.015 m
Y = 80 mm = 0.08 m; x = 120 mm = 0.12 m,

N_1 = 290 rpm or $\omega_1 = \dfrac{2\pi \times 290}{60} = 30.4 rad/\sec$

N_2 = 310 rpm $\omega_2 = \dfrac{2\pi \times 310}{60} = 32.5 rad/\sec$



(a) Lowest position

Fig. 6.

Problem 1. contd…

1. *Loads on the spring at the lowest and highest equilibrium speeds:*

Let S1 = spring load at lowest equilibrium speed
S2 = spring load at highest equilibrium speed.
Since the ball arms are parallel to governor axis at the lowest equilibrium speed (*i.e.* at *N*1 = 290 rpm), as shown in Fig. 6.
So, r = r_1 = 120 mm = 0.12 m

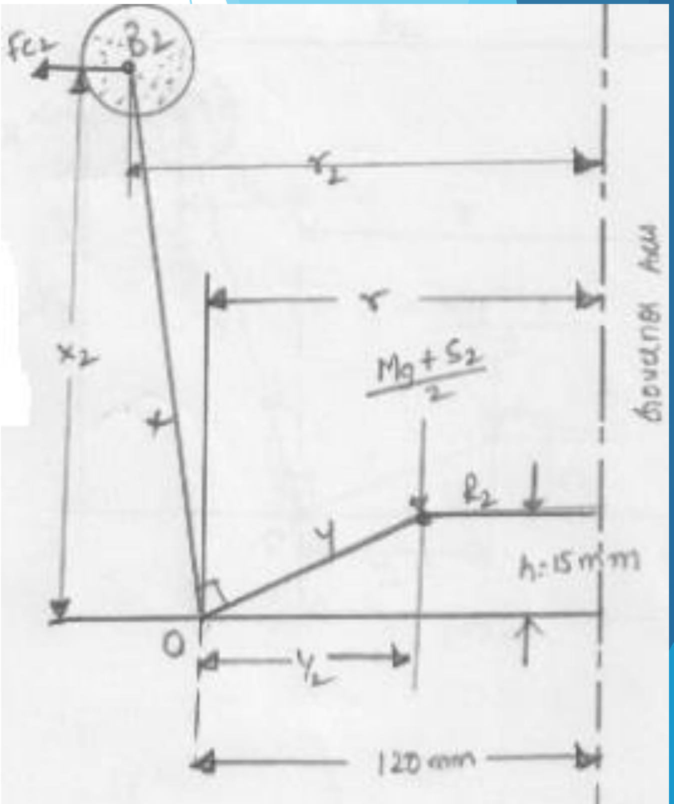
Centrifugal force at the minimum speed:

$$F_{C1} = m_1.\omega_1^2.r_1 = 2.5\times(30.4)^2\times 0.12 = 277\,N$$

The **position of ball arm and sleeve arm at the highest equilibrium speed** is shown in Fig. 7.

$$h = (r_2 - r_1)\frac{y}{x}$$

$$r_2 = r_1 + \frac{h.x}{y} = 0.12 + \frac{0.015\times 0.12}{0.08} = 0.1425m$$



(b) Highest position

Fig. 7.

Problem 1. contd…

Centrifugal force at the maximum speed:

$$F_{C2} = m_2.\omega_2^2.r_2 = 2.5\times(32.5)^2\times 0.1425 = 376N$$

Neglecting the obliquity effect of arms and the moment due to the weight of the balls, we have for lowest position:

$$M.g + S_1 = 2F_{C1}\frac{x}{y} = 2\times 277\times\frac{0.12}{0.08} = 831N; at \quad M = 0$$

S_1 = 831 N

$$M.g + S_2 = 2F_{C2}\frac{x}{y} = 2\times 376\times\frac{0.12}{0.08} = 1128N; at \quad M = 0$$

S_2 = 1128 N

2. Stiffness of the spring:

$$s = \frac{S_2 - S_1}{h} = \frac{1128 - 831}{15} = 19.8N/mm$$

Thank You